



OPEN Large language models for zero-shot procedure extraction in orthopedic surgery: a comparative evaluation

Ashton Williamson¹, Nazgol Tavabi¹, Nishita Kalepalli¹, Ophelie Lavoie-Gagne¹, Andre Weiss¹, Benjamin Owens², Andrew Sibley², Rafael A. García Andújar¹, Alexandra Santos¹, Alexander Kim¹, Mahad M. Hassan^{1,2,3} & Ata M. Kiapour¹✉

Operative notes in electronic health records contain critical information for understanding surgical care, yet manual coding is time-consuming, costly, and inconsistent. Large language models (LLMs) promise to transform this process by automatically extracting detailed procedure information - a capability with significant implications for scaling clinical registries and advancing surgical research. Here, we conducted a large-scale evaluation of state-of-the-art LLMs for zero-shot structured information extraction from orthopedic clinical notes. Fourteen open-source and proprietary models were tested on 800 real operative notes, annotated by both an orthopedic surgeon and an administrator using a curated list of 74 procedure classes. We compared model outputs to human annotations, assessing accuracy and exploring the effects of model scale, reasoning capabilities, and prompt design. We find that across models, LLMs consistently outperform administrator-assigned labels, achieving macro-F1 scores above 0.6 and improving over administrative coding by up to 10 points. Larger models and reasoning capabilities further boosted performance, though gains plateaued beyond 30 billion parameters. Performance varied by procedure frequency, revealing clear strengths and persistent challenges for rare or complex cases. Modern LLMs can already outperform routine administrative coding in extracting detailed surgical procedure data, pointing to a future where registry curation could be faster, cheaper, and more consistent. Yet, full alignment with surgical experts remains an open challenge- especially for rare procedures - emphasizing the need for domain adaptation and thoughtful deployment. Our findings illustrate how general-purpose LLMs can advance automated clinical data curation and inform the next generation of surgical informatics.

Clinical registries, also referred to as patient or disease registries, are organized repositories of patient data that are collected through observational means for an express purpose¹. The construction of clinical registries has emerged as a critical task in healthcare with the potential to enhance clinical research, improve patient quality of care and treatment outcomes, and reduce the cost burden of healthcare^{2–5}. Despite these benefits, the deployment of clinical registries is often hindered by the high costs associated with implementation and maintenance. In a recent survey of organizations operating clinical registries, 64% listed cost as a barrier to sustainability while another survey of high-risk device registries found that only half of respondents received sustainable registry funding^{6,7}. One potential avenue for mitigating the high financial cost of registry construction is the use of secondary data sources, particularly data captured within the electronic health record (EHR) system. While the process of transforming EHR data to fit a registry schema is straightforward for structured data, it is significantly more challenging for unstructured data, like clinical notes. Transforming unstructured clinical notes into variables of interest in the registry is a task that has traditionally necessitated skilled human abstractors, in a process commonly referred to as chart review. Machine-learning-based natural language processing (NLP) algorithms have provided a promising alternative to this paradigm by improving the cost burden, turnaround time, and overall accuracy but are often limited in practice by the need for annotated, in-domain training data^{8–11}. Recent work has highlighted the potential of large language models (LLMs) to further expedite the

¹Department of Orthopedics and Sports Medicine, Boston Children's Hospital, Harvard Medical School, Boston, MA 02115, United States. ²Department of Orthopedic Surgery, University of Minnesota, Minneapolis, MN 55454, United States. ³Department of Orthopedic Surgery, TRIA Orthopaedic Center, Bloomington, MN 55122, United States. ✉email: ata.kiapour@childrens.harvard.edu

registry creation process by circumventing the need for tailored training data^{12–15}. Questions remain, however, as to the robustness of these approaches to task complexity, label space dimensionality, variations in medical subdomains and other implementation challenges.

In this work we investigate the extent to which LLMs can assist in the creation and curation of clinical registries by evaluating their ability to extract clinically significant variables from unstructured medical notes in a zero-shot setting. We construct a dataset consisting of operative notes drawn from a high-volume hip-preservation clinic that treats children and young adults, a setting that introduces additional heterogeneity in diagnoses, operative techniques, and documentation styles. We select this domain deliberately because hip surgeries are both common and clinically complex, often involving several different simultaneous procedures. While recent studies have applied large language models to either procedural coding or free-text information extraction (for example, named-entity recognition), these approaches often fall short for large-scale clinical analysis because raw extracted spans can be inconsistent, ambiguous, or difficult to aggregate. In practice, downstream applications such as registries require consistent, standardized representations of complex procedures. This need is especially acute in domains like orthopedic hip surgery, where common CPT codes may group together distinct operations – such as open femoral head/neck osteochondroplasty, open labral repair, periacetabular osteotomy, and triple innominate osteotomy – under the same code, obscuring important procedural differences¹⁶. To address this, we framed the task not as open-ended extraction, but as multi-label classification into a carefully defined set of orthopedic procedure classes. This approach bridges the gap between LLMs' ability to parse rich procedural detail and the practical requirement for structured, analyzable data. This presents a departure from previous work in LLM-based procedure classification from orthopedic operative notes, which focused solely on CPT-code classification for a set spinal surgery related CPT codes¹⁷. Our contributions include the following:

1. We perform a large-scale evaluation of state-of-the-art large language models (LLMs) for zero-shot multi-label classification of orthopedic operative notes to support clinical registry construction.
2. We develop and utilize an expert-annotated dataset covering 74 procedure classes across 800 real-world operative notes.
3. We demonstrate that modern LLMs consistently outperform administrator-assigned procedure labels, with the best model improving macro-F1 by 0.10.
4. We analyze the effect of model scaling on performance and efficiency and show diminishing returns beyond 30 billion parameters, while highlighting the benefit of reasoning capabilities ("thinking tokens") and the impact of different prompting strategies.
5. We leverage our pipeline to construct a comprehensive, 20-year registry of over 80,000 hip procedures across thousands of pediatric patients, which is currently supporting multiple ongoing clinical research efforts.

Results

Dataset

The test set consisted of 800 operative notes, each from a distinct patient and hospital encounter, from 2000 to 2023. Average note length was approximately 1057 tokens (using the GPT-4o tokenizer) with a standard deviation of 913 tokens. In the expert-annotated evaluation set, 90% of the procedure classes (66/74) occur at least once, with an average of 2.64 procedure classes per operative note (± 2.15) and a maximum of 13 procedure classes assigned to a single note. We calculate Krippendorff's Alpha for the expert-annotated and administrator-annotated evaluation sets, which we find to be $\alpha = 0.714$.

Zero shot multilabel procedure classification

Multilabel classification on the expert annotated set was run across all the models in Table 2. In order to compare the performance across different models we ranked them in terms of F1-score using a Critical Difference (CD) Diagram on the top 25 most frequent classes¹⁸, demonstrated in Fig. 1. Ranks and significance bars in the CD Diagram are computed using the Friedman test with a post hoc Wilcoxon signed-rank test with Holm's Correction. Full results of all models across all metrics can be found in Supplementary Materials Appendix F. Based on Fig. 1, the best performing model on the expert evaluation set is GPT-o4-mini, achieving an accuracy of 0.986, a macro-averaged F1-score of 0.624, and a macro-averaged Matthews Correlation Coefficient of 0.637. Additionally, GPT-o4-mini outperforms the admin set by 0.09 in micro-averaged F1-score and 0.10 in macro-averaged F1-score. It is also the only LLM that showed a significant difference compared to the administrator

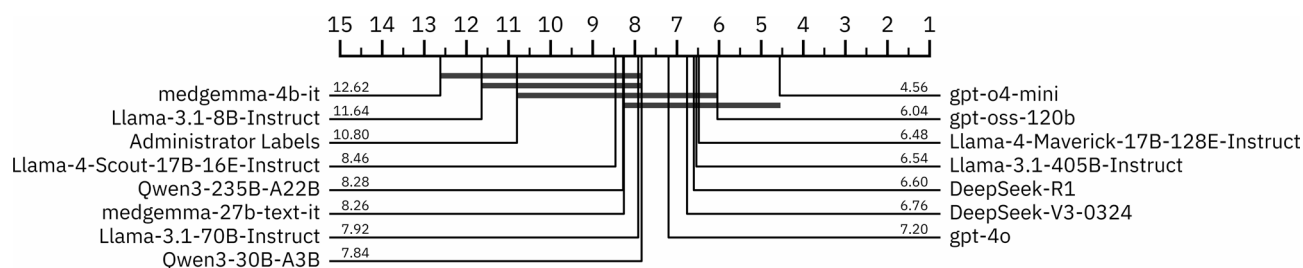


Fig. 1. Critical decision diagram ranking evaluated approaches in terms of F1 Score on the top 25 most frequent classes. Horizontal lines denote approaches with no significant difference.

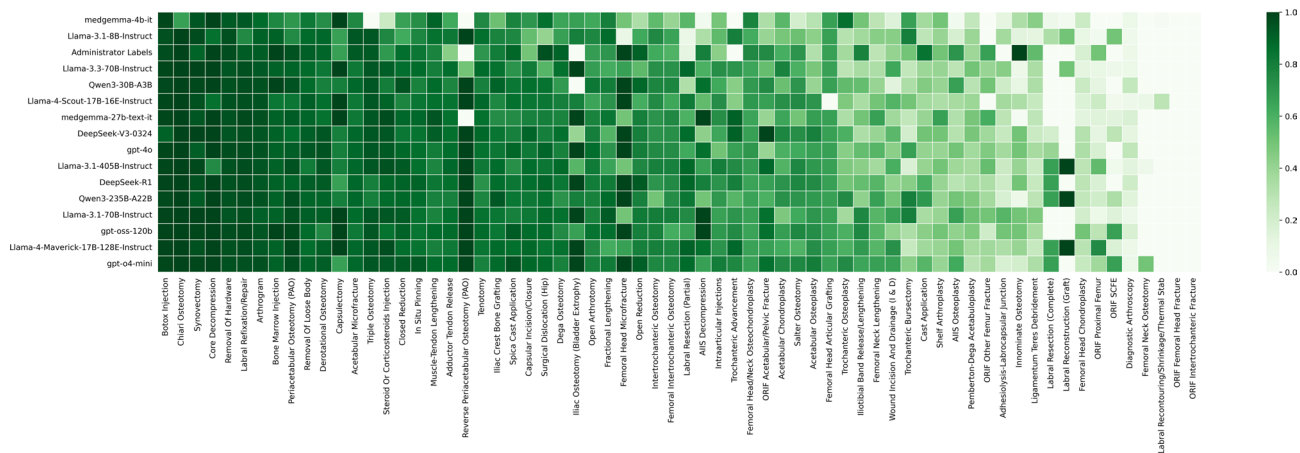


Fig. 2. Heatmap of F1-score for each approach evaluated against the expert-annotated set. Procedures are ordered by decreasing mean F1 score.

	Frequency	F1-Score		MCC	
		GPT-o4-mini	Admin Set	GPT-o4-mini	Admin Set
Innominate Osteotomy	1	0.200	1.000	0.332	1.000
ORIF Femoral Head Fracture	1	0.000	0.000	0.000	−0.001
ORIF Intertrochanteric Fracture	1	0.000	0.000	−0.001	0.000
ORIF SCFE	1	0.667	0.000	0.707	−0.002
Labral Reconstruction (graft)	1	0.000	0.000	0.000	0.000
Femoral Neck Osteotomy	1	0.500	0.000	0.577	−0.003
Femoral head Microfracture	1	1.000	1.000	1.000	1.000
Shelf Arthroplasty	1	0.400	0.500	0.499	0.577
Reverse PAO	1	1.000	0.000	1.000	0.000
Chiari Osteotomy	2	1.000	1.000	1.000	1.000
Iliac Crest Bone Grafting	74	0.883	0.809	0.871	0.800
Tenotomy	75	0.790	0.710	0.768	0.705
Open Arthrotomy	75	0.793	0.677	0.778	0.669
Arthrogram	82	0.963	0.925	0.959	0.917
Spica Cast Application	84	0.954	0.893	0.949	0.883
Periacetabular Osteotomy	85	0.965	0.952	0.961	0.947
Intraarticular Injections	86	0.805	0.435	0.784	0.435
Removal of Hardware	128	0.959	0.965	0.952	0.959
Botox Injection	178	0.989	0.971	0.986	0.964
Capsular Incision/Closure	192	0.885	0.504	0.849	0.510

Table 1. Per-class F1-score and Matthew’s Correlation Coefficient for GPT-o4-mini and administrator-assigned labels on the top 10 most common and top 10 least common procedure classes.

annotated labels ($p = 0.00091$), as indicated in the CD diagram in Fig. 1. The administrator labels, in contrast, are ranked lower than 12 out of 14 models on the top 25 classes, though they are not significantly different from the other seven worst-performing models.

We provide a heatmap of F1-scores across all models and procedure classes in Fig. 2. As shown in Fig. 2, GPT-o4-mini achieves consistently higher F1-scores across procedure classes. In contrast, the administrator-annotated set shows more variable performance. We additionally compare per-class results in terms of F1-score between the best-performing model on the expert-annotated set and the administrator-annotated set (Table 1 presents the top and bottom 10 procedure classes by frequency, with the full set of results provided in Table 7 in Appendix F of the Supplementary Materials). GPT-o4-mini outperforms the administrator-assigned labels in F1-score for the majority of procedure classes with at least one positive label (43 out of 66). This performance gap is more pronounced among frequently occurring classes – defined as those above the median frequency – where GPT-o4-mini outperforms in 76% of cases, compared to 55% for rarer classes.

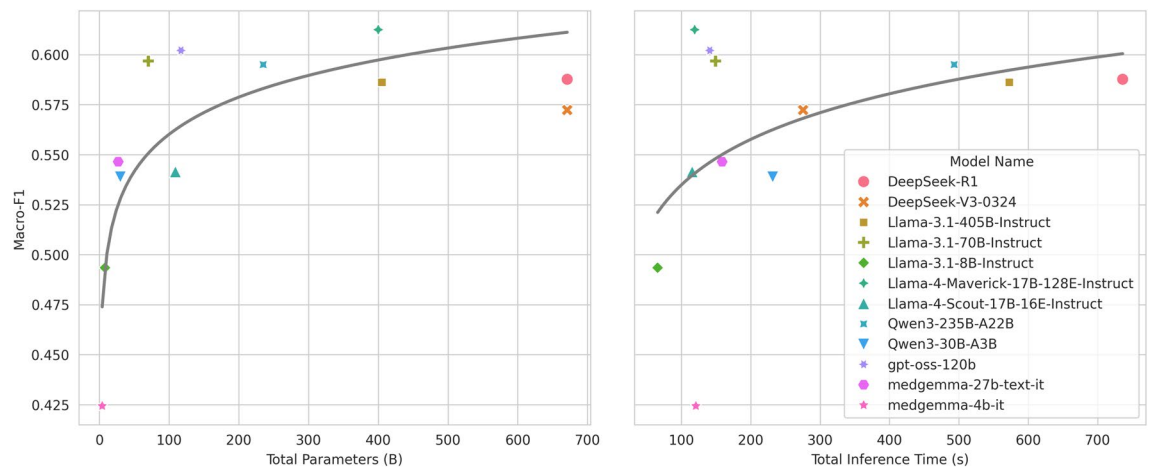


Fig. 3. Macro-F1 as a function of total model parameters (in billions of parameters) and total inference time on the evaluation set (in seconds). Regression lines are created using ordinary least squares in logarithm of x-axis.

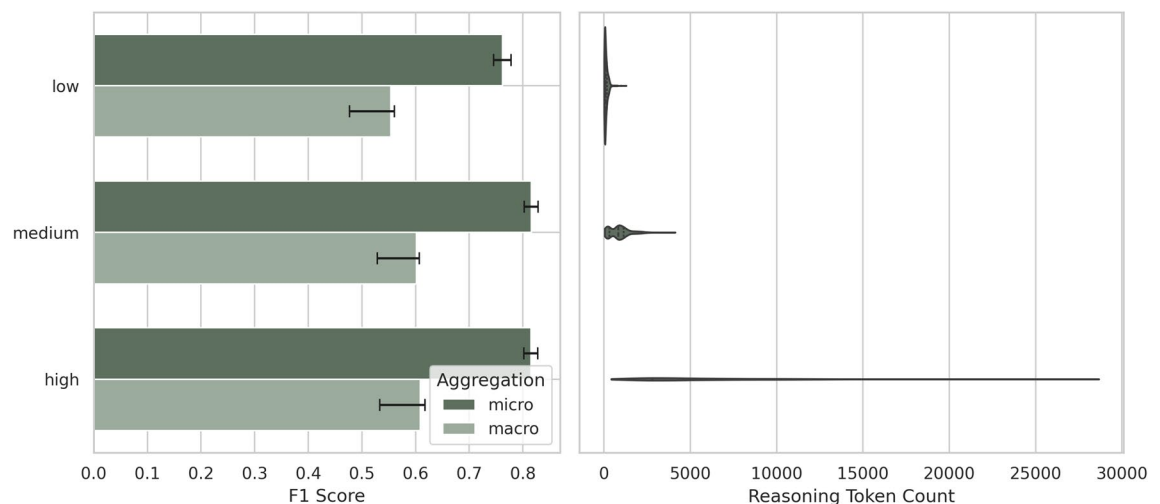


Fig. 4. F1 score (both micro- and macro-averaged) on the evaluation set by reasoning level (left) is contrasted against the corresponding distribution of reasoning response length in tokens as a violin plot (right). Error bars in the F1 score plot represent a 95% bootstrap confidence interval (9,999 replications).

Model comparison and performance analysis

We find an overall positive correlation between model size, measured in total parameters, and micro-averaged F1-score on the expert-annotated evaluation set. We note however, diminishing returns for models larger than 30B parameters (Fig. 3). Despite the general trend of higher performance resulting from larger models, we do note some scaling disparity between different families of models. We also observe that models that use thinking tokens perform better than their non-thinking counterparts. In particular, DeepSeek R1 (reasoning) outperforms DeepSeek V3 (non-reasoning) on the expert-annotated set in terms of both micro- and macro-F1 (+0.03 and +0.02, respectively) and GPT-o4-mini outperforms the non-reasoning GPT-4O by 0.02 micro-F1 and 0.04 macro-F1. These improvements, however, fail to account for differences in training setups between the models. For a less biased comparison, we test the Qwen3 reasoning models (Qwen3-235B-A22B and Qwen3-30B-A22B) with and without thinking tokens enabled. We provide the full Reasoning Ablations results in Supplementary Materials Appendix K. This trend holds for the larger of the two models, Qwen3-235B-A22B, which sees a modest decrease in F1 after disabling thinking tokens. In contrast, the smaller model sees negligible performance degradation in terms of micro-F1 (< 0.01) and an increase in macro-F1 after disabling thinking tokens. We note, however, that this performance increase comes at a cost of throughput, with reasoning models taking significantly longer to process the full evaluation set (see Fig. 3). Figure 4 shows the result of varying reasoning effort for GPT-OSS using the predefined reasoning level parameters (*low*, *medium*, and *high*). We see a increase in macro-averaged F1 score when transitioning from *low* to *medium* reasoning level (+4.8%, absolute) at the cost of an additional 750 reasoning tokens per response, on average. Increasing reasoning token budget

beyond the *medium* reasoning effort level results in negligible returns in macro-averaged F1 score— +0.7% absolute increase from *medium* to *high* reasoning effort— while significantly increasing average response length (+4841 tokens per response on average). This translates to a total run time on the evaluation set of 83 seconds at *low* reasoning effort, 133 seconds at *medium*, and 583 at *high*. Furthermore, we observe that models pretrained on biomedical text (MedGemma) outperform identical general-domain models (Gemma). This effect is even more pronounced for smaller models, with MedGemma-4B seeing a more than two-fold increase over Gemma-4B in both micro-averaged and macro-averaged F1-score (see Appendix H).

Prompting strategies

All models evaluated using the single-label classification prompt saw a decrease in micro-averaged F1-score. We also note a general increase in total positive predictions, resulting in a higher micro-averaged recall for all models, though this ultimately outweighed by a comparatively larger decrease in micro-averaged precision. We see similar F1-score performance degradation when classifying extracted procedure sections as opposed to the full note. The exception to this was for the smallest model tested, at 8 billion parameters, which saw a slight increase in F1-score (+3.3%). In contrast to single-label classification, however, all models evaluated on procedure span classification exhibit higher micro-averaged recall but lower micro-averaged precision. The final multi-label classification prompt, its refinement process, and the impact of tailored class instructions on performance are provided in Appendix E.

Discussion

The extraction of structured data elements from unstructured text within the Electronic Health Record System is a critical task with the potential to enhance the precision of clinical decision-making by enabling large-scale analyses of patient data. While large language models have gained popularity on the task of structured data retrieval in the medical domain^{19–21}, the extent to which models can replace expert annotation remains unclear, particularly in specialized sub-domains for which benchmark data is scarce. In this study we investigated the alignment of large language models with human annotators of varying levels of expertise on the task of classifying hip procedures performed in orthopedic hip operations using associated operative notes. Our approach was later used to build a registry of over 20 years of data, capturing 86,555 distinct hip procedures across more than 12,000 patients, with an average age of 14 (See Appendix L in Supplementary Materials for more details). This process completed in under just two hours using GPT-OSS-120B, compared to our human annotators for which the same task took weeks for a dataset of less than 3% of the size.

Our annotation strategy used two complementary annotator roles: an administrator annotator and a pediatric orthopedic hip preservation expert. This design reflects a clinically realistic abstraction workflow, in which administrative personnel may perform initial data extraction while domain experts provide reference interpretation for specialized clinical classification tasks. However, each annotator role was represented by a single annotator. As a result, although we can compare administrator and expert labels, we cannot fully characterize within-role variability or estimate how labels might vary across multiple independent experts. This limitation is relevant for operative-note classification, where ambiguous procedure descriptions may support more than one plausible label. Given the specialized expertise required and the time-intensive nature of annotation, additional independent expert annotation was not feasible in this study. Future work with multiple expert annotators across institutions would further evaluate annotation variability and generalizability.

While GPT-o4-mini demonstrates the best performance on the expert annotated set, the difference between the top-performing models across all procedure classes is not statistically significant and a number of smaller, open-source models perform comparably. Among these models, we note three factors that generally lead to enhanced performance: model size as measured by total parameters, test-time compute scaling with thinking tokens, and training data tailored to the medical domain. While the positive association between F1 performance and total parameters is not unexpected, it is worth noting the diminishing returns in model size. Similarly, models that utilize reasoning capabilities or are trained primarily on medical data tend to demonstrate better performance on the evaluation set. This comes at the cost of additional output tokens (*reasoning* tokens) resulting in decreased overall throughput and increased financial cost if using the model through a paid API with a token-based pricing model. Improving performance through opting for reasoning models or increasing the reasoning token budget thus becomes an obvious choice in deployment environments where registry data is processed in an offline setting— in which inference is batched and latency is not a concern— and where the LLM is self-hosted, in order to mitigate the financial penalty incurred by the longer responses.

All approaches demonstrate a high level of per-procedure performance variability, excelling at identifying some procedures while consistently struggling on others. While some of this inter-class performance disparity is model dependent, we find some procedures to be consistently challenging for all LLMs. One example is the set of open reduction and internal fixation (ORIF) procedures, for which the mean macro-F1 across models was less than 0.25. We find that this is most often due to ambiguous references to femoral fracture locations in the notes (which are necessary for classification of 4 out of 7 of these procedures) as well as inconsistent documentation of ORIF procedures by surgeons. We also note the challenge of distinguishing between similar class names for different procedures— for example, *periacetabular osteotomy* and *acetabular osteoplasty*. We find that providing the entire list of classes is necessary to allow the LLM to discriminate between multiple similar classes, as evidenced by the decrease in precision resulting from providing individual classes for classification in isolation. Despite the zero-shot nature of the prompting strategy, all models exhibit at least a slight positive correlation between F1-score and procedure frequency in the expert-annotated test set (see Appendix G for a plot of Matthew's Correlation Coefficient vs Label Frequency). Although we suspect that this discrepancy can be contributed to either shortcomings of the label taxonomy or annotation bias, we do not discard the possibility

that the LLMs may have had more exposure to the more common procedures in their training corpus, leading to an increased familiarity with these classes.

While even state-of-the-art models failed to achieve high alignment with our expert annotator on the evaluation set, we emphasize the importance of evaluating against annotators with less expertise, given the practical realities of chart review implementation. To this end, we find that state-of-the-art, proprietary LLMs outperform the administrative annotator when evaluating against an expert annotator on the task of orthopedic hip procedure classification. Perhaps more surprising, however, is that all but two out of thirteen models outperformed the administrative annotator on the expert set. This includes models with just 27 billion parameters, or approximately 4% of the size of the largest open-source LLM tested.

These findings support the potential utility of LLMs in clinical information extraction and, ultimately, in creating and maintaining clinical registries. The minor relative difference in metric performance between the top open weight models and state of the art proprietary models further suggests the viability of efficient, on-premises deployment alternatives for clinical information extraction. Nonetheless, challenges remain in incorporating these systems into real-world clinical workflows. For one, it can be challenging to quantify uncertainty in model predictions, making it difficult to identify samples or individual predictions in need of further manual review. While this may hinder integration into human-in-the-loop systems, it does not entirely preclude it. As human-assistance tools in the clinical information extraction process, LLMs enable systems for online annotation suggestions as well as retroactive error identification—both essential methods for streamlining registry curation. Likewise, ethical concerns in unsupervised LLM deployment remain a principal dilemma. Given the lack of explainability of these systems, comprehensive evaluation and continued oversight are paramount to ensure equitable operation.

There are a few key limitations to this study. We first note the relatively small size of the dataset and the lack of multi-institutional data, which both serve to limit the generalizability of our results. Additionally, we emphasize that both our expert-annotated set and our administrator-annotated set are annotated by individual annotators as opposed to multiple annotators, which we highlight as a fundamental limitation of this study. Finally, we note the inherent limitations of assigning a discrete taxonomy to complex and varied medical procedures. We conjecture that more well-defined rules for procedure assignment and more transparent class definitions would lead to better replicability in the annotation process.

Methods

Dataset

The present study was reviewed by the Boston Children's Hospital Institutional Review Board, who determined that it met the criteria for exemption and waived the requirement of obtaining informed consent (IRB-P00046914). All methods were conducted in accordance with relevant guidelines and regulations. Following approval, we obtained notes from all Boston Children's Hospital clinics spanning 2000–2023. Using a supervised machine learning pipeline developed in-house⁸, we filter out operative notes to include only hip-related operations. This process yielded 30,192 operative notes for 12,203 unique patients across 22,415 encounters. In parallel, with input from a team of orthopedic surgeons, we manually curated a detailed set of common hip procedures with focus on ensuring uniqueness, relevance, and comprehensive coverage. We first constructed a candidate set of procedures using procedure data from internally-maintained surgical registries. This set was reviewed independently by a team of hip surgeons, who proposed additional procedure classes as well as omissions of obsolete procedures and consolidations of functionally identical procedures. The resulting set was edited to maintain consistent naming conventions. The final set comprised 74 distinct surgical procedures (the full list of procedures is provided in Appendix A of Supplementary Materials).

Due to the expected long-tail distribution of our labels and the strong effect of sampling on evaluation outcomes^{22–24}, we developed a targeted approach to increase representation of less common classes in the evaluation set. We first embed all documents using doc2vec²⁵, then cluster using k-means (with k equal to the number of desired samples), and, finally, sample a single document from within each cluster with probability proportional to their distance from the centroid (for more information on the sampling method, see Supplementary Materials Appendix B). Using this method, we selected 800 operative notes for manual annotation (dataset and sampled-set information, including counts, patient demographics, and note characteristics, are provided in Table 3 of Appendix C in the Supplementary Materials). To assess model alignment with domain experts of varying experience, the sampled notes were annotated once by an orthopedic surgeon and once by an orthopedic administrator with over 10 years of experience, including in conducting chart review. Annotators were each tasked with assigning binary procedure labels to each of the operative notes in the dataset, corresponding to whether the procedure class was conducted as a procedure in the described operation. A visualization of areas of agreement and disagreement across procedure labels between annotators is provided in Figure 4 in Appendix C of Supplementary Materials. All annotations were completed using a custom-built, in-house annotation application developed specifically with this study in mind to facilitate efficient and accurate annotation of operative notes (see Appendix D in Supplementary Materials for details).

Zero shot multilabel procedure classification

Given an operative note, the goal is to classify the procedures in the operative note into any of the procedure classes without access to external knowledge sources or reference examples. Models are provided with a description of the task along with a list of the procedure classes and prompted to return a JSON object containing a key for each of the procedures and a binary value corresponding to whether a procedure belonging to the class occurred in the operation. Model output is parsed to generate a label vector, where each element corresponds to the presence of a particular procedure from the ordered list of candidate procedures. We experiment with several different open-source LLMs as well as two closed-source, proprietary LLMs (see Table 2). All open-source LLMs

Model	Params	Active Params	Reasoning	Open Weights
Llama-3.1-8B-Instruct ²⁶	8B	-	✗	✓
Llama-3.1-70B-Instruct ²⁶	70B	-	✗	✓
Llama-3.1-405b-Instruct ²⁶	405B	-	✗	✓
medgemma-4b-it ²⁷	4B	-	✗	✓
medgemma-27b-text-it ²⁷	27B	-	✗	✓
Qwen3-30B-A3B ²⁸	30B	3B	✓	✓
Qwen3-235B-A22B ²⁸	235B	22B	✓	✓
DeepSeek-R1 ²⁹	671B	37B	✓	✓
DeepSeek-V3-0324 ³⁰	671B	37B	✗	✓
Llama-4-Scout-17B-16E-Instruct ³¹	109B	17B	✗	✓
Llama-4-Maverick-17B-128E-Instruct ³¹	400B	17B	✗	✓
gpt-oss-120B ³²	117B	5B	✓	✓
gpt-o4-mini-2025-04-16	-	-	✓	✗
gpt-4o-2024-11-20	-	-	✗	✗

Table 2. Overview of large language models tested on the hip procedure evaluation dataset. For Mixture of Experts models, we list the number of active parameters, where available. Reasoning Models denote models that support some form of “thinking” tokens to scale test time compute.

are run using the VLLM library on eight NVIDIA H200 GPUs using default sampling parameters suggested by the respective model authors. Proprietary OpenAI models are accessed through a HIPAA-secure institutional API. Models are evaluated against the annotated evaluation set using macro- and micro-averaged Matthew's Correlation Coefficient, F1, and accuracy. Given the long-tail distribution of procedures in our evaluation dataset, we additionally provide an analysis of per-class performance.

Model comparison and performance analysis

The proliferation of both open-source and proprietary LLMs and the saturation of many common benchmarks has come to complicate the process of choosing an LLM, particularly in lower-resource settings. To assess the extent to which model architecture and design choices affect performance on our dataset, we evaluate the impact of model size (measured in total model parameters), use of reasoning tokens, and differences in computation time across models. Additionally, given the recent success of reasoning models, a term which we use to refer to models that leverage thinking tokens in their responses³³, we evaluate the extent to which reasoning ability improves performance on our expert-annotated test set. We evaluate both the relative performance difference for reasoning models after disabling thinking tokens as well as the performance differences in similarly sized reasoning and non-reasoning models. To measure the effect of reasoning budget, we employ the open-weight reasoning model from OpenAI— GPT-OSS-120B— that supports dynamically changing the average reasoning effort per response (token length) by modifying the system prompt to include one of three predefined levels: low, medium (the default) and high. This allows for effectively varying the reasoning budget while still allowing the model to terminate its chain of thought organically, without forced stopping. Finally, we evaluate the effect of increased exposure to medical data during model training by comparing the performance of two models of identical architecture trained on general domain (Gemma) and biomedical datasets (MedGemma).

Prompting strategies

Given the potentially long operative notes and the large number of procedure labels (> 70), both the input and expected output for the LLMs can be extensive. Since LLMs have been shown to be susceptible to perturbations in prompt setup³⁴, we tested different prompting strategies and measured their impact on task performance. The first of these was single-label classification, in which the LLM is provided a full operative note and a single procedure to classify, instead of the full list of procedures (for a comparison of the prompts, see Fig. 5). While multilabel classification is a more intuitive approach to the task, it is possible that the complexity introduced by the multilabel prompting setup could adversely affect performance. Providing individual classes, in comparison, may be a more manageable task for an LLM.

The second approach was procedure span classification, in which the structured *procedures performed* field is used in place of the entire document. In the US, operative notes tend to adhere to a template with predefined fields for procedural details that are required for billing and reporting. This includes, at minimum, patient information, surgery date and time, names of surgical staff, pre- and post-op diagnoses, procedure(s) performed, anesthesia type, complications, implants, and procedural details³⁵. For the task of procedure classification, two of these sections appear to be the most relevant: procedures performed and procedural details. The former of these is a more structured field containing a free-text list of procedures performed (e.g. “1. Removal of hardware, left hip 2. Wound irrigation”) while the latter contains a detailed, long-form description of the specifics of the operation (an example de-identified operative note is provided in Fig. 2 of Appendix C in the Supplementary Materials). For all other experiments, the full content of the operative note is provided to the LLM as context. To test the extent to which the *procedures performed* section could act as a stand in for the full text, we extracted the *procedures performed* field from the operative note and provided it to the LLM in place of the full text. In effect,

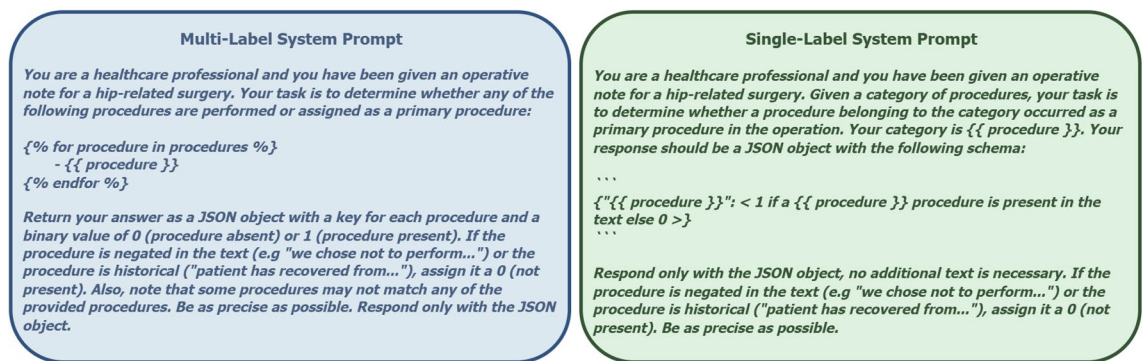


Fig. 5. A comparison of system prompts used for multi-label (our default task prompt, left) and single-label (right) classification experiments. The *procedure(s)* variable is inserted dynamically at runtime.

this approach both reduces overall input length and reduces the noise introduced by the full body of the operative note, providing a brief clear summary of the operation performed, at the expense of comprehensiveness. Full details of these approaches are provided in the Supplementary Materials Appendices I and J.

Data availability

The data used in this study cannot be publicly shared due to patient privacy concerns. Data can be accessed upon reasonable request pending approvals from Boston Children's Hospital.

Received: 17 March 2026; Accepted: 8 May 2026

Published online: 21 May 2026

References

- Gliklich, R. E., Leavy, M. B. & Dreyer, N. A. Registries for Evaluating Patient Outcomes: A User's Guide en. Tech. rep. Edition: Fourth edition (Agency for Healthcare Research and Quality (AHRQ), Sept. 2020). <https://effectivehealthcare.ahrq.gov/products/registries-guide-4th-edition/users-guide> (2025).
- Stey, A. M. et al. Clinical registries and quality measurement in surgery: A systematic review. *Surgery* 157, 381–395. issn: 0039–6060. <https://www.sciencedirect.com/science/article/pii/S0039606014006072> (2025) (Feb. 2015).
- Larsson, S., Lawyer, P., Garellick, G., Lindahl, B. & Lundström, M. Use of 13 disease registries in 5 countries demonstrates the potential to use outcome data to improve health care's value. *Health Aff.* 31, 220–227. <https://doi.org/10.1377/hlthaff.2011.0762> (2025).
- Hoque, D. M. E. et al. Impact of clinical registries on quality of patient care and clinical outcomes: A systematic review. en. PLOS ONE 12. Publisher: Public Library of Science, e0183667. issn: 1932–6203. <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0183667> (2025) (Sept. 2017).
- Lee, P. et al. Economic evaluation of clinical quality registries: a systematic review. en. BMJ Open 9, e030984. issn: 2044–6055, 2044–6055. <https://bmjopen.bmj.com/lookup/doi/10.1136/bmjopen-2019-030984> (2025) (Dec. 2019).
- Blumenthal, S. The Use of Clinical Registries in the United States: A Landscape Survey. eGEMs 5, 26. issn: 2327–9214. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5994955/> (2025) (2017).
- Becherer, B. E. et al. High risk device registries: Global value, costs, and sustainable funding. *J. Plast. Reconstr. Aesthet. Surg.* 71, 1362–1380 (2018).
- Tavabi, N. et al. Building large-scale registries from unstructured clinical notes using a low-resource natural language processing pipeline. *Artificial Intelligence in Medicine* 151, 102847. issn: 0933–3657. <https://www.sciencedirect.com/science/article/pii/S0933365724000897> (2025) (May 2024).
- Tavabi, N., Singh, M., Pruneski, J. & Kiapour, A. M. Systematic evaluation of common natural language processing techniques to codify clinical notes. *PLoS One* 19, e0298892. <https://doi.org/10.1371/journal.pone.0298892> (2025).
- Tavabi, N. et al. Disparities in cannabis use and documentation in electronic health records among children and young adults. en. npj Digital Medicine 6. Publisher: Springer Science and Business Media LLC. issn: 2398–6352. <https://www.nature.com/articles/s41746-023-00885-w> (2025) (Aug. 2023).
- Murphy, R. M. et al. Adverse drug event detection using natural language processing: A scoping review of supervised learning methods. en. PLOS ONE 18. Publisher: Public Library of Science, e0279842. issn: 1932–6203. <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0279842> (2025) (Jan. 2023).
- Liu, L. et al. Using natural language processing to extract information from clinical text in electronic medical records for populating clinical registries: A systematic review. *J. Am. Med. Inform. Assoc.* 33, 484–499 (2026).
- Sakai, H. & Lam, S. S. Large language models for health care text classification: Systematic review. *JMIR AI* 5, e79202 (2026).
- Hu, Y. et al. Information extraction from clinical notes: Are we ready to switch to large language models?. *J. Am. Med. Inform. Assoc.* <https://doi.org/10.1093/jamia/ocaf213> (2026).
- Gu, B. et al. Scalable information extraction from free text electronic health records using large language models. *BMC Med. Res. Methodol.* 25, 23 (2025).
- Spence, D. D. et al. Coding Challenges in Common Pediatric and Adolescent Hip Preservation Procedures. *Journal of the Pediatric Orthopaedic Society of North America* 4, 471. issn: 2768–2765. <https://www.sciencedirect.com/science/article/pii/S2768276524006175> (2025) (May 2022).
- Zaidat, B. et al. Artificially intelligent billing in spine surgery: An analysis of a large language model. *Glob. Spine J.* 15, 1113–1120 (2025).
- Fawaz, H. I., Forestier, G., Weber, J., Idoumghar, L. & Muller, P.-A. Deep learning for time series classification: a review. *Data Mining and Knowledge Discovery* 33. [arXiv:1809.04356](https://arxiv.org/abs/1809.04356) [cs], 917–963. issn: 1384–5810, 1573–756X. <https://link.springer.com/article/10.1007/s10618-019-00619-1> (2025) (July 2019).

19. Azar, W. S. et al. LLM-Mediated Data Extraction from Patient Records after Radical Prostatectomy. *NEJM AI* 2. Publisher: Massachusetts Medical Society, A1cs2400943. <https://ai.nejm.org/doi/full/10.1056/A1cs2400943> (2025) (May 2025).
20. Ferber, D. et al. GPT-4 for Information Retrieval and Comparison of Medical Oncology Guidelines. *NEJM AI* 1. Publisher: Massachusetts Medical Society, A1cs2300235. <https://ai.nejm.org/doi/full/10.1056/A1cs2300235> (2025) (May 2024).
21. Unlu, O. et al. Retrieval-augmented generation-enabled GPT-4 for clinical trial screening. *NEJM AI* 1, A1oa2400181 (2024).
22. Roberts, M. et al. Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. en. *Nature Machine Intelligence* 3. Publisher: Nature Publishing Group, 199–217. issn: 2522–5839. <https://www.nature.com/articles/s42256-021-00307-0> (2025) (Mar. 2021).
23. Sechidis, K., Tsoumakas, G. & Vlahavas, I. On the Stratification of Multi-label Data en. in *Machine Learning and Knowledge Discovery in Databases* (eds Gunopulos, D., Hofmann, T., Malerba, D. & Vazirgiannis, M.) (Springer, Berlin, Heidelberg, 2011), 145–158. isbn: 978-3-642-23808-6.
24. Merrillees, M. & Du, L. Stratified Sampling for Extreme Multi-label Data en. in *Advances in Knowledge Discovery and Data Mining* ISSN: 1611–3349 (Springer, Cham, 2021), 334–345. isbn:978-3-030-75765-6. https://link.springer.com/chapter/10.1007/978-3-030-75765-6_27 (2025).
25. Le, Q. & Mikolov, T. Distributed representations of sentences and documents in International conference on machine learning. 1188–1196 (2014).
26. Dubey, A. et al. The llama 3 herd of models. arXiv e-prints, [arXiv: 2407.21783](https://arxiv.org/abs/2407.21783). (2024).
27. Sellergren, A. et al. MedGemma Technical Report. July 2025. [arXiv:2507.05201](https://arxiv.org/abs/2507.05201) [cs]. (2025).
28. Yang, A. et al. Qwen3 Technical Report. May 2025. [arXiv:2505.09388](https://arxiv.org/abs/2505.09388)[cs] (2025).
29. DeepSeek-AI et al. DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning Jan. 2025. [arXiv:2501.12948](https://arxiv.org/abs/2501.12948) [cs]. (2025).
30. DeepSeek-AI et al. DeepSeek-V3 Technical Report Feb. 2025. [arXiv:2412.19437](https://arxiv.org/abs/2412.19437) [cs]. (2025).
31. The Llama 4 herd: The beginning of a new era of natively multimodal AI innovation en. <https://ai.meta.com/blog/llama-4-multimodal-intelligence/> (2025).
32. Agarwal, S. et al. gpt-oss-120b & gpt-oss-20b Model Card 2025. [arXiv:2508.10925](https://arxiv.org/abs/2508.10925) [cs.CL].
33. Jaech, A. et al. OpenAI o1 System Card 2024. [arXiv:2412.16720](https://arxiv.org/abs/2412.16720) [cs.AI].
34. Bommasani, R. et al. On the Opportunities and Risks of Foundation Models. July 2022. [arXiv:2108.07258](https://arxiv.org/abs/2108.07258) [cs]. (2025).
35. Centers for Medicare & Medicaid Services. State Operations Manual, Appendix A: Survey Protocol, Regulations and Interpretive Guidelines for Hospitals Rev. 238. See 42 CFR §482.51(b)(6), Surgical Services, Tag A-0959, operative report requirements. Centers for Medicare & Medicaid Services (2026). https://www.cms.gov/regulations-and-guidance/guidance/manuals/downloads/som107_ap_a_hospitals.pdf.

Acknowledgements

We would like to thank the following individuals for their assistance with this study: Shefali R. Bijwadia, Harsev Singh, Joseph Murray, Ariana Goli, and Leili Sarmadi.

Author contributions

AMK was responsible for securing funding for this study. AWilliamson, NT, NK, MH, and AMK contribution to the development of methodology and conducted the investigation. NT and MH provided project administration and supervision, with additional supervision from AWilliamson. Data curation, validation, and visualization were performed by AWilliamson, NK, OL, AWeiss, BO, RG, ASantos, ASibley and AK. AWilliamson developed the codebase, conducted formal analyses, and prepared the original manuscript draft. AWilliamson, NT, NK, OL, AW, BO, RG, ASantos, ASibley, AK, and MH contributed to critical review and editing.

Funding

NVIDIA Applied Research Accelerator, Oracle Research, Children's Hospital Orthopaedic Surgery Foundation, and the Oberg Family Endowment.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1038/s41598-026-52928-3>.

Correspondence and requests for materials should be addressed to A.M.K.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026